

INDIA BIODIVERSITY MAP

NATIONAL PARKS, WILDLIFE SANCTUARIES & BIODIVERSITY HOTSPOTS

IMPORTANT NATIONAL PARKS

- 1 Jim Corbett NP (Uttarakhand)
Tiger, Elephant
- 2 Ranthambore NP (Rajasthan)
Tiger
- 3 Gir NP (Gujarat)
Asiatic Lion
- 4 Kaziranga NP (Assam)
One-horned Rhino
- 5 Sundarbans NP (West Bengal)
Royal Bengal Tiger
- 6 Kanha NP (Madhya Pradesh)
Barasingha (Swamp Deer)
- 7 Bandhavgarh NP (Madhya Pradesh)
Tiger
- 8 Periyar NP (Kerala)
Elephant

IMPORTANT WILDLIFE SANCTUARIES

- 1 Keoladeo Ghana NP (Bharatpur, Rajasthan)
Migratory Birds
- 2 Periyar WLS (Kerala)
Elephant
- 3 Bhadra WLS (Karnataka)
Tiger, Elephant
- 4 Nalsarovar WLS (Gujarat)
Flamingo, Water Birds
- 5 Chilika WLS (Odisha)
Migratory Birds

LEGEND

- 1 National Parks
- 3 Wildlife Sanctuaries
- Forest Cover
- Himalaya Hotspot
- Western Ghats – Sri Lanka Hotspot



BIODIVERSITY HOTSPOTS IN INDIA

- **Himalaya Hotspot**
(J&K, Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh)
- **Western Ghats – Sri Lanka Hotspot**
(Gujarat, Maharashtra, Goa, Karnataka, Kerala, Tamil Nadu)

MAJOR SPECIES & THEIR HABITAT

-  **Royal Bengal Tiger**
Sundarbans (WB), Corbett (UK), Ranthambore (RJ), Kanha (MP)
-  **Asiatic Lion**
Gir (Gujarat)
-  **One-horned Rhino**
Kaziranga (Assam)
-  **Elephant**
Western Ghats (Kerala, Karnataka, Tamil Nadu), Assam, Odisha
-  **Snow Leopard**
Himalayan Region (J&K, HP)
-  **Gharial**
Chambal River (MP, Rajasthan)

BIOSPHERE RESERVES IN INDIA

- Nilgiri Biosphere Reserve – Tamil Nadu, Kerala, Karnataka
- Nanda Devi Biosphere Reserve – Uttarakhand
- Gulf of Mannar Biosphere Reserve – Tamil Nadu
- Great Nicobar Biosphere Reserve – Andaman & Nicobar Islands
- Sunderbans Biosphere Reserve – West Bengal



NOTE: India is one of the 17 Mega-Diverse Countries in the world with rich flora and fauna.



DAY 1 – LATITUDE, LONGITUDE & IMPORTANT LINES

1. Latitude lines are imaginary horizontal lines running parallel to the Equator and measure distance north or south.
2. Longitude lines are vertical lines that measure distance east or west of the Prime Meridian.
3. The Equator (0° latitude) divides Earth into Northern and Southern Hemispheres.
4. The Prime Meridian (0° longitude) passes through Greenwich and divides Earth into Eastern and Western Hemispheres.
5. Tropic of Cancer (23.5°N) passes through India and marks northern limit of direct sunlight.
6. Tropic of Capricorn (23.5°S) marks southern limit of direct sunlight.
7. Arctic Circle (66.5°N) and Antarctic Circle (66.5°S) define polar regions.
8. Latitude affects climate, while longitude is important for time calculation.
9. Earth rotates 360° in 24 hours, so 15° longitude = 1 hour time difference.
10. Indian Standard Time is based on 82.5°E longitude.
11. Time zones are created to maintain uniform time in regions.
12. Parallels decrease in size from equator to poles, while meridians converge at poles.

DAY 2 – MOTIONS OF EARTH

1. Rotation is Earth's spinning on its axis, causing day and night.
2. Earth completes one rotation in 24 hours.
3. Revolution is Earth's movement around the Sun in an elliptical orbit.
4. One revolution takes 365¼ days, causing leap year every 4 years.
5. Earth's axis is tilted at 23.5°, causing seasons.

6. Summer solstice (21 June) has longest day in Northern Hemisphere.
7. Winter solstice (22 December) has shortest day in Northern Hemisphere.
8. Equinoxes (21 March, 23 September) have equal day and night.
9. Seasons are opposite in Northern and Southern Hemispheres.
10. Rotation causes Coriolis force affecting wind direction.
11. Revolution affects length of day and seasons.

DAY 3 – INTERIOR OF EARTH + ROCKS

1. Earth has three layers: crust, mantle, and core.
2. Crust is the outermost layer and supports life.
3. Mantle is the thickest layer with semi-solid magma.
4. Core is made of iron and nickel and is very hot.
5. Rocks are classified into igneous, sedimentary, and metamorphic.
6. Igneous rocks are formed from cooling magma.
7. Sedimentary rocks are formed by deposition of sediments.
8. Metamorphic rocks are formed by heat and pressure.
9. Rock cycle explains transformation of rocks from one type to another.
10. Minerals are building blocks of rocks.

DAY 4 – VOLCANOES, EARTHQUAKES & PLATE TECTONICS

1. Earth's crust is divided into tectonic plates.
2. Plate movement causes earthquakes and volcanoes.

3. Divergent boundaries create new crust.
4. Convergent boundaries cause subduction and mountain formation.
5. Transform boundaries cause earthquakes.
6. Earthquakes occur due to sudden release of energy.
7. Focus is origin point, epicenter is surface point.
8. Volcanoes erupt magma, gases, and ash.
9. Pacific Ring of Fire is most active zone.
10. India lies in seismic zones III–V.

DAY 5 – GEOMORPHOLOGY (LANDFORMS)

1. Landforms are shaped by internal and external forces.
2. Rivers form V-shaped valleys, meanders, and deltas.
3. Glaciers form U-shaped valleys and moraines.
4. Wind forms dunes and loess deposits.
5. Sea waves form cliffs, caves, arches, and stacks.
6. Erosion, transportation, and deposition shape landforms.
7. Floodplains are fertile regions.
8. Oxbow lakes form from cut-off meanders.
9. Coastal landforms change continuously.
10. Wind action is strong in deserts.

DAY 8–9 – ATMOSPHERE & WINDS

1. Atmosphere has layers: troposphere, stratosphere, mesosphere, thermosphere.
2. Weather occurs in troposphere.
3. Ozone layer is in stratosphere.
4. Pressure differences create winds.

5. Trade winds, westerlies, and polar winds are global winds.
6. Coriolis force deflects winds.
7. ITCZ is low-pressure zone near equator.
8. Jet streams influence weather.
9. Insolation controls temperature.
10. Heat budget maintains balance.

DAY 10 – INDIAN MONSOON

1. Monsoon is seasonal reversal of winds.
2. SW monsoon brings rainfall to India.
3. NE monsoon brings rain to Tamil Nadu.
4. Arabian Sea and Bay branches distribute rain.
5. ITCZ shifts northward in summer.
6. El Niño weakens monsoon.
7. La Niña strengthens monsoon.
8. Monsoon has breaks.
9. Western Ghats cause orographic rainfall.
10. Himalayas block cold winds.

DAY 11 – CYCLONES & OCEAN CURRENTS

1. Cyclones form over warm oceans.
2. Eye is calm center.
3. Bay of Bengal cyclones are stronger.
4. Warm currents increase rainfall.
5. Cold currents reduce rainfall.
6. Gulf Stream warms Europe.
7. Ocean currents affect climate.
8. Upwelling creates fishing grounds.
9. El Niño affects global weather.
10. Cyclones rotate anticlockwise in Northern Hemisphere.

DAY 12 – TIDES & OCEAN RELIEF

1. Tides are caused by Moon's gravity.
2. Spring tides occur during full moon.
3. Neap tides have low range.
4. Waves transfer energy.
5. Tsunami is caused by earthquakes.
6. Continental shelf is shallow region.
7. Abyssal plains are flat.
8. Trenches are deepest parts (Mariana Trench).
9. Mid-ocean ridges form new crust.
10. Shelf areas are rich in fish.

DAY 14 – PHYSICAL FEATURES OF INDIA IN

1. India has 6 physical divisions.
2. Himalayas are young fold mountains.
3. Northern plains are fertile.
4. Plateau is oldest landmass.
5. Thar desert is arid region.
6. Western coast is narrow.
7. Eastern coast is wide.
8. Islands include Andaman & Lakshadweep.
9. Mount Everest is highest peak.
10. Rivers shape plains.

DAY 15 – INDIAN CLIMATE

1. India has tropical monsoon climate.
2. Latitude, altitude affect climate.
3. Himalayas influence rainfall.
4. Monsoon dominates rainfall.
5. Western disturbances bring winter rain.

6. Seasons: winter, summer, monsoon, retreating.
7. Rainfall uneven distribution.
8. Coastal areas have moderate climate.
9. Interior regions have extreme climate.
10. Monsoon variability affects agriculture.

DAY 16 – RIVERS OF INDIA

1. Rivers are Himalayan and Peninsular.
2. Himalayan rivers are perennial.
3. Peninsular rivers are seasonal.
4. Ganga River is most important river.
5. Brahmaputra River is largest system.
6. Narmada River flows west.
7. East rivers form deltas.
8. West rivers form estuaries.
9. Rivers support agriculture.
10. Dams are built on rivers.

DAY 17 – SOIL, VEGETATION, AGRICULTURE

1. Soil types: alluvial, black, red, laterite, desert.
 2. Alluvial soil is most fertile.
 3. Black soil is good for cotton.
 4. Vegetation depends on climate.
 5. Evergreen forests get heavy rainfall.
 6. Deciduous forests shed leaves.
 7. Kharif crops grow in monsoon.
 8. Rabi crops grow in winter.
 9. Zaid crops grow in summer.
 10. Agriculture depends on monsoon.
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DAY 18 – MINERALS & INDUSTRIES

1. Minerals are natural resources.
 2. Types: metallic, non-metallic, energy.
 3. Iron and coal are important.
 4. Petroleum found in Assam & Gujarat.
 5. Industries depend on raw materials.
 6. Steel industry is backbone.
 7. IT industry growing fast.
 8. Resources are renewable/non-renewable.
 9. Conservation is important.
 10. Industrial location factors vary.
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DAY 22 – ECOSYSTEM

1. Ecosystem = biotic + abiotic.
 2. Producers make food.
 3. Consumers depend on others.
 4. Decomposers recycle nutrients.
 5. Energy flow is unidirectional.
 6. 10% law applies.
 7. Food chain transfers energy.
 8. Food web stabilizes ecosystem.
 9. Nutrient cycles exist.
 10. Balance is important.
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DAY 23 – POLLUTION

1. Pollution harms environment.
2. Air pollution causes diseases.
3. Water pollution affects life.
4. Soil pollution reduces fertility.
5. Noise pollution affects health.
6. Smog occurs in cities.

7. Acid rain damages crops.
 8. Ozone depletion increases UV.
 9. Eutrophication reduces oxygen.
 10. Control measures needed.
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DAY 24 – NATIONAL PARKS

1. Protected areas conserve biodiversity.
2. Jim Corbett National Park = tiger.
3. Gir National Park = lion.
4. Kaziranga National Park = rhino.
5. Sundarbans National Park = mangrove.
6. Sanctuaries allow limited use.
7. Biosphere reserves protect ecosystems.
8. India is mega-diverse country.
9. Mapping is important.
10. Conservation programs exist.

1. ORIGIN OF THE EARTH

1. The origin of the Earth is explained through different theories, among which the Nebular Hypothesis states that the solar system formed from a rotating cloud of gas and dust.
 2. According to the Big Bang Theory, the universe began about 13.8 billion years ago from a highly dense and hot singularity, which later expanded to form galaxies and stars.
 3. The solar system, including Earth, is believed to have formed around 4.6 billion years ago from a collapsing nebula.
 4. Planetesimal theory suggests that small particles collided and stuck together due to gravity, gradually forming larger bodies like planets.
 5. The Earth initially existed in a molten state, which gradually cooled to form a solid crust over millions of years.
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2. STRUCTURE OF THE EARTH (INTERIOR)

1. The Earth is composed of three main layers: crust, mantle, and core, each having distinct physical and chemical properties.
 2. The crust is the outermost layer of the Earth, which is very thin and ranges from about 5 km under oceans to about 70 km under continents.
 3. The continental crust is mainly composed of silica and aluminum and is therefore called “SIAL.”
 4. The oceanic crust is mainly composed of silica and magnesium and is therefore called “SIMA.”
 5. The mantle lies beneath the crust and extends up to a depth of about 2900 km, making it the thickest layer of the Earth.
 6. The mantle is composed mainly of silicate rocks rich in iron and magnesium and behaves like a semi-solid material.
 7. The core is the innermost layer and is primarily composed of iron and nickel, hence it is called “NIFE.”
 8. The outer core is in a liquid state, while the inner core is solid due to extremely high pressure.
 9. The temperature and pressure increase as we go deeper into the Earth’s interior.
 10. The core is responsible for generating the Earth’s magnetic field.
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3. HOW DO WE KNOW ABOUT EARTH’S INTERIOR?

1. Direct sources like mining and drilling provide limited information because humans have only reached a few kilometers below the surface.
2. Indirect sources such as seismic waves generated during earthquakes provide the most important information about Earth’s interior.
3. There are two main types of seismic waves: P-waves (primary waves) and S-waves (secondary waves).
4. P-waves can travel through solids, liquids, and gases, while S-waves can travel only through solids.
5. The absence of S-waves in certain layers indicates the presence of liquid material, such as the outer core.

4. IMPORTANT CONCEPTS (EXAM BOOSTER)

1. The boundary between crust and mantle is called the Moho (Mohorovičić discontinuity).
 2. The boundary between mantle and core is called the Gutenberg discontinuity.
 3. The lithosphere includes the crust and the uppermost part of the mantle and is broken into tectonic plates.
 4. The asthenosphere lies below the lithosphere and is partially molten, allowing plate movements.
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REVISION (QUICK RECALL)

- Earth = Crust + Mantle + Core
- Core = NIFE (Nickel + Iron)
- Crust = SIAL (continental), SIMA (oceanic)
- P-wave = travels everywhere
- S-wave = only solids

MEMORY TRICKS

- **SIAL = SI + AL → Silica + Aluminium (Continents)**
- **SIMA = SI + MA → Silica + Magnesium (Oceans)**
- **NIFE = NI + FE → Nickel + Iron (Core)**

DAY 2 – LATITUDES, LONGITUDES & IMPORTANT LINES

1. LATITUDES (PARALLELS OF LATITUDE)

1. Latitudes are imaginary horizontal lines drawn parallel to the Equator that measure the angular distance of a place north or south of the Equator.
2. The Equator is the 0° latitude and divides the Earth into the Northern Hemisphere and Southern Hemisphere.
3. Latitudes range from 0° at the Equator to 90° North at the North Pole and 90° South at the South Pole.
4. All latitudes are parallel to each other and never intersect, which is why they are called parallels.
5. The length of latitudes decreases as we move from the Equator towards the poles.
6. Latitudes are used to determine climatic zones because temperature varies mainly with distance from the Equator.

2. IMPORTANT LATITUDES (VERY IMPORTANT FOR EXAM)

1. The Tropic of Cancer lies at 23.5° North latitude and passes almost through the middle of India.
2. The Tropic of Capricorn lies at 23.5° South latitude and marks the southern boundary of the tropical zone.
3. The Arctic Circle lies at 66.5° North latitude and marks the beginning of the polar region in the Northern Hemisphere.
4. The Antarctic Circle lies at 66.5° South latitude and marks the polar region in the Southern Hemisphere.
5. The region between Tropic of Cancer and Tropic of Capricorn is called the Tropical Zone, which receives maximum heat.

6. The regions beyond the Arctic and Antarctic Circles are called Frigid Zones, which are extremely cold.

3. LONGITUDES (MERIDIANS OF LONGITUDE)

1. Longitudes are imaginary vertical lines that run from the North Pole to the South Pole and measure angular distance east or west of the Prime Meridian.
2. The Prime Meridian is 0° longitude and passes through Greenwich in London.
3. Longitudes range from 0° to 180° East and 0° to 180° West.
4. Unlike latitudes, all longitudes are equal in length and converge at the poles.
5. Longitudes are used to determine time because the Earth rotates from west to east.

4. TIME & LONGITUDE (VERY IMPORTANT CONCEPT)

1. The Earth rotates 360° in 24 hours, which means it rotates 15° in one hour.

REVISION (QUICK RECALL)

- Latitude = Horizontal (Climate)
- Longitude = Vertical (Time)
- $15^\circ = 1$ hour
- IST = 82.5° E
- Equator = 0° latitude
- Prime Meridian = 0° longitude

2. Therefore, the time difference between two longitudes of 15° is one hour.
3. The Greenwich Mean Time (GMT) is the standard time at 0° longitude.
4. India follows Indian Standard Time (IST), which is based on 82.5° East longitude.
5. IST is 5 hours 30 minutes ahead of GMT because $82.5^\circ \div 15^\circ = 5.5$ hours.

5. IMPORTANT CONCEPTS (EXAM BOOSTER)

1. The Equator is the longest latitude and divides the Earth into two equal halves.
2. The Prime Meridian divides the Earth into Eastern and Western Hemispheres.
3. The International Date Line is located approximately at 180° longitude and is used to change the date.
4. Crossing the International Date Line from west to east subtracts one day, while crossing from east to west adds one day.

MEMORY TRICKS

- Latitude = “Ladder” (horizontal steps)
- Longitude = “Long lines” (vertical pole to pole)
- $15^\circ = 1$ hour $\rightarrow$ remember “15 = 1 time shift”
- India Time = $82.5^\circ \rightarrow “8 + 2.5 = 10.5 \rightarrow$ think 5.5 hrs ahead”

DAY 3 – MOTIONS OF THE EARTH

1. ROTATION OF THE EARTH

1. Rotation refers to the movement of the Earth on its axis, which is an imaginary line passing through the North Pole and South Pole.
 2. The Earth rotates from west to east, which is why the Sun appears to rise in the east and set in the west.
 3. One complete rotation takes approximately 24 hours, which results in the formation of day and night.
 4. The speed of rotation is highest at the Equator and decreases towards the poles.
 5. Rotation causes the deflection of winds and ocean currents due to the Coriolis effect.
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2. REVOLUTION OF THE EARTH

1. Revolution refers to the movement of the Earth around the Sun in an elliptical orbit.
 2. One complete revolution takes about 365 days and 6 hours, which is why a leap year is added every four years.
 3. The Earth's axis is tilted at an angle of 23.5° , which is the main reason for seasonal changes.
 4. The distance between Earth and Sun changes during revolution, but seasons are mainly caused by axial tilt, not distance.
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3. SEASONS (VERY IMPORTANT)

5. EQUINOXES

Spring (Vernal) Equinox – 21 March

1. On this day, the Sun is directly overhead at the Equator.
2. Day and night are equal all over the Earth.
3. It marks the beginning of spring in the Northern Hemisphere.

1. Seasons are caused due to the combined effect of Earth's revolution around the Sun and the tilt of its axis.
 2. When the Northern Hemisphere is tilted towards the Sun, it experiences summer, while the Southern Hemisphere experiences winter.
 3. When the Northern Hemisphere is tilted away from the Sun, it experiences winter, while the Southern Hemisphere experiences summer.
 4. The variation in day length and angle of sunlight causes temperature differences across seasons.
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4. SOLSTICES

Summer Solstice (21 June)

1. On this day, the Sun is directly overhead at the Tropic of Cancer (23.5° N).
 2. It is the longest day and shortest night in the Northern Hemisphere.
 3. The Northern Hemisphere experiences maximum sunlight and summer season.
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Winter Solstice (22 December)

1. On this day, the Sun is directly overhead at the Tropic of Capricorn (23.5° S).
 2. It is the shortest day and longest night in the Northern Hemisphere.
 3. The Northern Hemisphere experiences winter season.
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Autumnal Equinox – 23 September

1. On this day, the Sun is again directly overhead at the Equator.
2. Day and night are equal everywhere.
3. It marks the beginning of autumn in the Northern Hemisphere.

6. IMPORTANT CONCEPTS (EXAM BOOSTER)

1. The tilt of Earth's axis (23.5°) is the most important factor responsible for seasons.
2. If there were no axial tilt, there would be no seasons on Earth.
3. The equinox is the only time when the entire Earth experiences equal day and night.
4. During solstices, one hemisphere gets maximum sunlight while the other gets minimum.

REVISION (QUICK RECALL)

- Rotation → Day & Night
- Revolution → Seasons
- Tilt = 23.5°
- 21 June → Longest day (North)
- 22 Dec → Shortest day (North)
- 21 March & 23 Sept → Equal day-night

MEMORY TRICKS

- **“June = Joy (long day)”** → 21 June longest day
- **“Dec = Dark (short day)”** → 22 Dec shortest day
- **Equinox = Equal (day = night)**

DAY 4 – INTERIOR OF EARTH + ROCKS & MINERALS

1. INTERIOR OF THE EARTH (DETAILED UNDERSTANDING)

1. The Earth's interior is divided into three major layers: crust, mantle, and core, based on composition and physical properties.
2. The crust is the outermost and thinnest layer, forming the surface on which we live and where all geological activities are visible.
3. The mantle lies beneath the crust and extends up to about 2900 km, and it is composed of dense silicate rocks rich in iron and magnesium.
4. The core is the innermost layer and is mainly composed of iron and nickel, which makes it extremely dense and hot.
5. The outer core is liquid in nature, while the inner core is solid due to extremely high pressure despite high temperature.
6. The temperature increases as we go deeper into the Earth, reaching up to 5000°C or more in the core.
7. The pressure also increases with depth, which affects the physical state of materials inside the Earth.

2. IMPORTANT LAYERS (EXAM FOCUS)

1. The lithosphere is the rigid outer layer consisting of the crust and the uppermost part of the mantle.

2. The asthenosphere lies below the lithosphere and is semi-molten, allowing tectonic plates to move over it.
3. The mesosphere refers to the lower mantle, which is more rigid compared to the asthenosphere.
4. The core is divided into the outer core (liquid) and inner core (solid), which is important for magnetic field generation.

3. DISCONTINUITIES (VERY IMPORTANT)

1. The Mohorovičić discontinuity (Moho) separates the crust from the mantle.
2. The Gutenberg discontinuity separates the mantle from the core.
3. These discontinuities are identified through changes in seismic wave velocities.

4. ROCKS – BASIC UNDERSTANDING

1. Rocks are naturally occurring solid masses composed of minerals that form the Earth's crust.
2. Rocks are classified into three main types: igneous, sedimentary, and metamorphic rocks.
3. The transformation of rocks from one type to another is called the rock cycle.

5. IGNEOUS ROCKS

1. Igneous rocks are formed from the cooling and solidification of molten magma or lava.
2. Intrusive igneous rocks form beneath the Earth's surface and cool slowly, resulting in coarse-grained texture.
3. Extrusive igneous rocks form on the surface and cool quickly, resulting in fine-grained texture.
4. Igneous rocks are also called primary rocks because they are the first to form.

5. Examples include granite (intrusive) and basalt (extrusive).

6. SEDIMENTARY ROCKS

1. Sedimentary rocks are formed by the accumulation and compaction of sediments over long periods.
2. These rocks often contain fossils, which provide information about past life forms.
3. Sedimentary rocks are usually layered and are also called stratified rocks.
4. Examples include sandstone, limestone, and shale.

7. METAMORPHIC ROCKS

1. Metamorphic rocks are formed when existing rocks are transformed due to heat, pressure, or chemical changes.
2. This process is called metamorphism, and it changes the structure and composition of rocks.
3. Metamorphic rocks can be formed from both igneous and sedimentary rocks.
4. Examples include marble (from limestone) and slate (from shale).

8. ROCK CYCLE (CONCEPTUAL)

1. The rock cycle is a continuous process through which rocks change from one type to another over geological time.
2. Igneous rocks can break down into sediments to form sedimentary rocks.
3. Sedimentary rocks can transform into metamorphic rocks under heat and pressure.
4. Metamorphic rocks can melt to form magma, which again solidifies into igneous rocks.

9. MINERALS

1. Minerals are naturally occurring inorganic substances with a definite chemical composition and physical properties.
 2. Rocks are made up of one or more minerals, which determine their characteristics.
 3. Common minerals include quartz, feldspar, mica, and iron ore.
 4. Minerals are important for industrial development and economic growth.
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10. IMPORTANT CONCEPTS (EXAM BOOSTER)

1. Igneous rocks do not contain fossils, while sedimentary rocks commonly contain fossils.
 2. Basalt is the most common rock of the oceanic crust.
 3. Granite is the most common rock of the continental crust.
 4. Metamorphic rocks are generally harder than their parent rocks.
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REVISION (QUICK RECALL)

- Crust → Thin
 - Mantle → Thickest
 - Core → NIFE
 - Igneous → From magma
 - Sedimentary → Layered + fossils
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- Metamorphic → Heat + pressure
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MEMORY TRICKS

- **IGNEOUS = IGNITION** (fire → magma)
 - **SEDIMENTARY = SETTLED** layers
 - **METAMORPHIC = “META = change”**
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DAY 5 – VOLCANOES, EARTHQUAKES & PLATE TECTONICS

1. PLATE TECTONICS (CORE CONCEPT)

1. The Earth's lithosphere is divided into several large and small tectonic plates that float over the semi-molten asthenosphere.
2. These plates are constantly moving due to convection currents in the mantle, which are driven by heat from the Earth's interior.
3. The movement of tectonic plates causes major geological activities such as earthquakes, volcanic eruptions, and mountain formation.
4. Plate tectonics theory replaced the earlier Continental Drift Theory by providing a scientific explanation for plate movements.

5. The boundaries between tectonic plates are the most active zones where most geological events occur.
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2. TYPES OF PLATE BOUNDARIES (VERY IMPORTANT)

➤ Divergent Boundary

1. Divergent boundaries are formed when two tectonic plates move away from each other.
2. Magma rises from below to fill the gap, creating new crust.
3. Mid-ocean ridges like the Mid-Atlantic Ridge are examples of divergent boundaries.

► Convergent Boundary

1. Convergent boundaries occur when two plates move towards each other.
 2. One plate may subduct under the other, leading to the formation of trenches and volcanoes.
 3. The Himalayas were formed due to the collision of the Indian Plate and Eurasian Plate.
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► Transform Boundary

1. Transform boundaries occur when two plates slide past each other horizontally.
 2. These boundaries do not create or destroy crust but cause earthquakes.
 3. The San Andreas Fault is a famous example.
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3. VOLCANOES

1. A volcano is an opening in the Earth's crust through which molten magma, gases, and ash are expelled.
 2. Volcanoes are mainly found along plate boundaries, especially in regions known as the "Ring of Fire."
 3. Lava is magma that reaches the Earth's surface during a volcanic eruption.
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TYPES OF VOLCANOES

1. Active volcanoes are those that have erupted recently or are likely to erupt again.
 2. Dormant volcanoes have not erupted for a long time but may erupt in the future.
 3. Extinct volcanoes are those that are not expected to erupt again.
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VOLCANIC LANDFORMS

1. Shield volcanoes are broad and gently sloping due to low-viscosity lava.
 2. Composite volcanoes are steep and formed by alternating layers of lava and ash.
 3. Volcanic plateaus are formed by extensive lava flows over large areas.
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4. EARTHQUAKES

1. An earthquake is the sudden shaking of the Earth's surface caused by the release of energy from within the Earth.
 2. This energy release occurs due to the movement of tectonic plates along faults.
 3. The point inside the Earth where the earthquake originates is called the focus (hypocenter).
 4. The point directly above the focus on the surface is called the epicenter.
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5. SEISMIC WAVES

1. Earthquakes generate seismic waves that travel through the Earth's interior and surface.
 2. Primary waves (P-waves) travel fastest and can pass through solids, liquids, and gases.
 3. Secondary waves (S-waves) travel only through solids and are slower than P-waves.
 4. Surface waves cause the most damage during an earthquake.
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6. MEASUREMENT OF EARTHQUAKES

1. The magnitude of an earthquake is measured using the Richter scale, which indicates the energy released.
 2. The intensity of an earthquake is measured using the Mercalli scale, which indicates the damage caused.
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7. IMPORTANT CONCEPTS (EXAM BOOSTER)

1. Most earthquakes and volcanoes occur along plate boundaries, especially in the Pacific Ring of Fire.
2. Convergent boundaries are associated with strong earthquakes and volcanic activity.
3. Divergent boundaries create new crust, while convergent boundaries destroy crust.
4. Earthquakes cannot be predicted accurately, but their impact can be reduced through preparedness.

REVISION (QUICK RECALL)

- Plates float on asthenosphere
- Divergent → Move apart
- Convergent → Collide
- Transform → Slide
- Volcano = magma eruption
- Earthquake = energy release

MEMORY TRICKS

- **Divergent = “Divide” (move apart)**
- **Convergent = “Come together”**
- **Transform = “Slide past”**
- **Focus = inside, Epicenter = above**

DAY 6 – GEOMORPHOLOGY (LANDFORMS)

1. GEOMORPHOLOGY (BASIC CONCEPT)

1. Geomorphology is the study of landforms and the processes that shape the Earth’s surface over time.
2. Landforms are created mainly by internal forces (endogenic) and external forces (exogenic).
3. Exogenic processes such as running water, glaciers, wind, and sea waves continuously modify the Earth’s surface.
4. The three major processes involved in landform formation are erosion, transportation, and deposition.

1. V-shaped valleys are formed in the upper course of rivers due to vertical erosion.
2. Waterfalls are formed when a river flows over a steep slope or hard rock surface.
3. Gorges and canyons are deep valleys with steep sides formed by continuous river erosion.

► DEPOSITIONAL LANDFORMS

1. Floodplains are flat areas formed by deposition of sediments during floods.
2. Meanders are bends in the river formed due to lateral erosion and deposition.
3. Oxbow lakes are formed when a meander is cut off from the main river.
4. Deltas are formed at river mouths where sediments are deposited into a sea or lake.

2. LANDFORMS BY RIVERS (FLUVIAL LANDFORMS)

► EROSIONAL LANDFORMS

3. LANDFORMS BY GLACIERS

► EROSIONAL LANDFORMS

1. Cirques are bowl-shaped depressions formed at the head of a glacier.
2. Arêtes are sharp ridges formed between two cirques.
3. U-shaped valleys are formed due to glacial erosion, which are wider and deeper than river valleys.

► DEPOSITIONAL LANDFORMS

1. Moraines are deposits of rocks and debris carried and deposited by glaciers.
2. Eskers are long, winding ridges formed by glacial deposits.
3. Drumlins are smooth, elongated hills formed by glacial action.

4. LANDFORMS BY WIND (AEOLIAN LANDFORMS)

► EROSIONAL LANDFORMS

1. Mushroom rocks are formed due to differential erosion by wind.
2. Yardangs are streamlined ridges formed by wind erosion.

► DEPOSITIONAL LANDFORMS

1. Sand dunes are formed by deposition of sand carried by wind.
2. Loess deposits are fine wind-blown sediments that form fertile soils.

REVISION (QUICK RECALL)

- River → V-shaped valley, delta
- Glacier → U-shaped valley, moraine
- Wind → dunes, loess

5. LANDFORMS BY SEA WAVES (COASTAL LANDFORMS)

► EROSIONAL LANDFORMS

1. Sea cliffs are steep rock faces formed due to wave erosion.
2. Sea caves are formed when waves erode weak parts of coastal rocks.
3. Sea arches are formed when caves enlarge and break through headlands.
4. Sea stacks are isolated rock columns left after arches collapse.

► DEPOSITIONAL LANDFORMS

1. Beaches are formed by deposition of sand and sediments along the coast.
2. Sandbars are ridges of sand formed parallel to the coast.
3. Lagoons are shallow water bodies separated from the sea by sandbars.

6. IMPORTANT CONCEPTS (EXAM BOOSTER)

1. Rivers mainly create V-shaped valleys, while glaciers create U-shaped valleys.
2. Deltas are formed in calm water conditions, while estuaries are formed in strong tidal conditions.
3. Wind action is most effective in desert regions due to lack of vegetation.
4. Coastal landforms are constantly changing due to wave action.

- Sea → cliffs, beaches

MEMORY TRICKS

- River = “V” shape

- **Glacier = “U” shape (like a bowl)**
- **Wind = Desert = Dunes**

- **Sea = Cliffs → Caves → Arches → Stacks (sequence)**

DAY 8–9 – ATMOSPHERE, WINDS, MONSOON & OCEAN CURRENTS

1. STRUCTURE OF THE ATMOSPHERE

1. The atmosphere is a layer of gases surrounding the Earth, which protects life and regulates temperature.
2. The troposphere is the lowest layer of the atmosphere, extending up to about 8 km at poles and 18 km at the equator, where all weather phenomena occur.
3. The stratosphere lies above the troposphere and contains the ozone layer, which absorbs harmful ultraviolet radiation.
4. The mesosphere lies above the stratosphere and is the coldest layer, where meteors burn upon entering the Earth’s atmosphere.
5. The thermosphere is the uppermost layer where temperatures are very high and auroras occur.

2. Not all incoming solar radiation is absorbed; some is reflected back into space by clouds, ice, and land surfaces.
3. The balance between incoming and outgoing radiation is called the heat budget of the Earth.
4. Temperature decreases with altitude in the troposphere at an average rate called the lapse rate.

3. PRESSURE BELTS & WINDS

PRESSURE BELTS

1. The Earth has alternating high and low pressure belts due to unequal heating of its surface.
2. The Equatorial region has low pressure due to rising warm air, while polar regions have high pressure due to cold descending air.

EXAM BOOSTER

- Weather = Troposphere
- Ozone layer = Stratosphere

2. HEAT BUDGET & TEMPERATURE

1. The Earth receives energy from the Sun in the form of solar radiation, which is called insolation.

WINDS (VERY IMPORTANT)

1. Winds are the horizontal movement of air from high-pressure areas to low-pressure areas.
2. Trade winds blow from subtropical high-pressure areas towards the equatorial low-pressure region.
3. Westerlies blow from subtropical regions towards subpolar low-pressure areas.

4. Polar easterlies blow from polar high-pressure areas towards subpolar regions.
5. The Coriolis force, caused by Earth's rotation, deflects winds to the right in the Northern Hemisphere and to the left in the Southern Hemisphere.

4. MONSOON SYSTEM (VERY IMPORTANT FOR INDIA)

1. Monsoon refers to seasonal reversal of winds, especially in the Indian subcontinent.
2. During summer, land heats up faster than the ocean, creating low pressure over land and drawing moist winds from the ocean.
3. This results in the Southwest Monsoon, which brings heavy rainfall to India.
4. During winter, land cools faster than the ocean, creating high pressure over land and causing dry winds to blow outward.
5. This results in the Northeast Monsoon, which brings rainfall mainly to Tamil Nadu.

EXAM BOOSTER

- SW Monsoon = Rain in most of India
- NE Monsoon = Rain in Tamil Nadu

5. CYCLONES (LINKED WITH CURRENT AFFAIRS)

1. Cyclones are low-pressure systems with strong winds rotating around a center.
2. Tropical cyclones form over warm ocean waters and require high temperature and moisture.

REVISION (QUICK RECALL)

- Troposphere → Weather
- Stratosphere → Ozone

3. In the Northern Hemisphere, cyclones rotate anticlockwise, while in the Southern Hemisphere, they rotate clockwise.

6. OCEAN CURRENTS

1. Ocean currents are continuous movements of ocean water caused by wind, temperature, salinity, and Earth's rotation.
2. Warm currents originate near the equator and move towards the poles, while cold currents originate near the poles and move towards the equator.
3. Ocean currents influence climate, especially coastal regions.

EL NIÑO & LA NIÑA (VERY IMPORTANT)

1. El Niño refers to the warming of surface waters in the Pacific Ocean, which weakens the Indian monsoon.
2. La Niña refers to the cooling of surface waters, which strengthens the Indian monsoon.

IMPORTANT CONCEPTS (EXAM BOOSTER)

1. Monsoon is driven by differential heating of land and sea.
2. Coriolis force affects wind direction but not speed.
3. Ocean currents regulate global climate patterns.
4. El Niño → weak monsoon, La Niña → strong monsoon.

-
- Winds → High → Low pressure
 - Monsoon → Seasonal wind reversal
 - El Niño → Weak monsoon

- La Niña → Strong monsoon

MEMORY TRICKS

- “Tropo = Weather”

- “Strato = Shield (Ozone)”
- “El Niño = No rain in India”
- “La Niña = Lots of rain”

DAY 10 – INDIAN MONSOON (ADVANCED) + CYCLONES + LOCAL WINDS

1. INDIAN MONSOON (ADVANCED CONCEPT)

1. The Indian Monsoon is a complex climatic system driven by differential heating of land and sea, pressure differences, and global atmospheric circulation.
2. During summer, intense heating over the Indian subcontinent creates a low-pressure area, which attracts moisture-laden winds from the Indian Ocean.
3. These winds are deflected due to the Coriolis force and become the Southwest Monsoon winds.
4. The monsoon is also influenced by the shifting of the Inter-Tropical Convergence Zone (ITCZ), which moves northward during summer.
5. The Tibetan Plateau plays a crucial role by heating up and strengthening the monsoon circulation.

2. BRANCHES OF SOUTHWEST MONSOON

Arabian Sea Branch

1. The Arabian Sea branch strikes the Western Ghats and causes heavy rainfall on the windward side due to orographic uplift.
2. The leeward side (rain shadow region) receives less rainfall.

Bay of Bengal Branch

1. The Bay of Bengal branch moves towards northeast India and causes heavy rainfall in regions like Assam and Meghalaya.
2. It then moves westward along the Himalayas, distributing rainfall across northern India.

3. WITHDRAWAL (RETREATING MONSOON)

1. The monsoon begins to retreat from northwest India in September due to decreasing temperature and pressure changes.
2. The retreating monsoon winds blow from land to sea and are dry in most parts of India.
3. However, these winds pick up moisture from the Bay of Bengal and cause rainfall in Tamil Nadu.

4. FACTORS AFFECTING MONSOON (VERY IMPORTANT)

1. El Niño tends to weaken the monsoon by reducing moisture availability.
2. La Niña strengthens the monsoon by increasing moisture and convection.

3. The Indian Ocean Dipole (IOD) also affects rainfall patterns in India.
4. Jet streams, especially the Tropical Easterly Jet, influence the onset and strength of monsoon.

5. CYCLONES (ADVANCED UNDERSTANDING)

1. Cyclones are intense low-pressure systems characterized by strong winds, heavy rainfall, and storm surges.
2. Tropical cyclones develop over warm ocean waters with temperatures above 27°C.
3. They require high humidity, low vertical wind shear, and a pre-existing disturbance to form.
4. Cyclones in the Bay of Bengal are generally more intense than those in the Arabian Sea due to higher sea surface temperatures.

TYPES OF CYCLONES

1. Tropical cyclones occur in low-latitude regions and are powered by heat from the ocean.
2. Temperate cyclones occur in mid-latitudes and are caused by temperature contrasts between air masses.

CYCLONE ROTATION

1. Cyclones rotate anticlockwise in the Northern Hemisphere due to the Coriolis force.

2. They rotate clockwise in the Southern Hemisphere.

6. LOCAL WINDS (VERY IMPORTANT FOR EXAMS)

Hot Winds

1. Loo is a hot and dry wind that blows in northern India during summer, especially in May and June.
2. Chinook is a warm, dry wind that blows down the eastern slopes of the Rocky Mountains and is known as the “snow eater.”
3. Foehn is a warm wind similar to Chinook, found in the Alps.

Cold Winds

1. Mistral is a cold, dry wind that blows from the Alps towards the Mediterranean region.
2. Bora is a cold wind that blows from the mountains towards the Adriatic Sea.

Moist Winds

1. Monsoon winds are seasonal winds that bring rainfall to India.
2. Sea breeze and land breeze are local winds caused by temperature differences between land and sea.

7. IMPORTANT CONCEPTS (EXAM BOOSTER)

1. Monsoon onset in India usually begins in Kerala around June 1.
2. The Western Ghats cause heavy rainfall due to orographic effect.
3. Tamil Nadu receives most of its rainfall from the Northeast Monsoon.
4. Cyclones are named differently in different regions (e.g., hurricanes, typhoons).

REVISION (QUICK RECALL)

- SW Monsoon → Summer rain

- NE Monsoon → Tamil Nadu rain
- Arabian branch → Western Ghats
- Bay branch → Northeast India
- El Niño → Weak monsoon
- Loo → Hot wind

MEMORY TRICKS

- “Arabian = Ghats rain”
- “Bay = Assam rain”
- “Loo = Heat wave in North India”
- “Chinook = Snow eater”

DAY 11 – CYCLONES (CASE STUDIES) + OCEAN CURRENTS (DETAILED)

1. CYCLONES (REVISION + ADVANCED LINK)

1. Cyclones are intense low-pressure systems that form over warm ocean waters and are characterized by strong winds, heavy rainfall, and storm surges.
2. They require sea surface temperatures above 27°C, high humidity, low vertical wind shear, and Coriolis force for development.
3. The eye of a cyclone is the calm center, while the eyewall is the most destructive region with maximum winds.
4. Cyclones weaken after landfall because they lose moisture supply from the ocean.

1. It struck Odisha in 2019 and was one of the strongest cyclones in recent years.
2. Efficient evacuation reduced the number of casualties significantly.
3. It is considered a successful example of disaster management in India.

2. CYCLONE CASE STUDIES (IMPORTANT FOR EXAM)

Cyclone Amphan

1. It was a super cyclone that hit eastern India, especially West Bengal and Odisha, in May 2020.
2. It caused massive destruction due to high wind speeds and storm surges.
3. It highlighted the importance of early warning systems and disaster preparedness.

Cyclone Tauktae

1. It formed in the Arabian Sea and affected western coastal states like Gujarat and Maharashtra.
2. It showed increasing cyclone frequency in the Arabian Sea due to warming oceans.

Cyclone Michaung

1. It affected Tamil Nadu and Andhra Pradesh with heavy rainfall and flooding.
2. It highlighted urban flooding issues in cities like Chennai.

Cyclone Fani

3. OCEAN CURRENTS (DETAILED CONCEPT)

1. Ocean currents are large-scale movements of ocean water driven by wind, temperature differences, salinity, and Earth's rotation.

2. Surface currents are mainly driven by winds, while deep ocean currents are driven by density differences (thermohaline circulation).
 3. Ocean currents play a major role in regulating global climate and influencing rainfall patterns.
-

4. TYPES OF OCEAN CURRENTS

Warm Currents

1. Warm currents originate near the equator and move towards higher latitudes.
 2. They increase the temperature of coastal regions and bring more rainfall.
-

Cold Currents

1. Cold currents originate near the poles and move towards the equator.
 2. They decrease coastal temperatures and reduce rainfall.
-

5. IMPORTANT OCEAN CURRENTS (MAP BASED)

Atlantic Ocean

1. Gulf Stream is a warm current that flows from the Gulf of Mexico towards Europe, warming the climate of Western Europe.
 2. Labrador Current is a cold current that flows from the Arctic towards eastern Canada.
-

Pacific Ocean

5.

REVISION (QUICK RECALL)

- Cyclone = Low pressure + warm ocean
- Eye = Calm center

1. Kuroshio Current is a warm current flowing along the east coast of Japan.
 2. Oyashio Current is a cold current flowing from the Arctic region.
-

Indian Ocean

1. Agulhas Current is a warm current flowing along the southeast coast of Africa.
 2. West Australian Current is a cold current flowing along the west coast of Australia.
-

6. EL NIÑO & LA NIÑA (REVISION + LINK)

1. El Niño is the abnormal warming of Pacific Ocean waters, which weakens Indian monsoon and reduces rainfall.
 2. La Niña is the abnormal cooling of Pacific Ocean waters, which strengthens monsoon and increases rainfall.
 3. These phenomena affect global weather patterns, including droughts and floods.
-

7. IMPORTANT CONCEPTS (EXAM BOOSTER)

1. Warm currents increase rainfall, while cold currents reduce rainfall in coastal regions.
 2. Regions where warm and cold currents meet are rich fishing grounds due to nutrient mixing.
 3. Cyclones mostly occur between 5°–20° latitude due to Coriolis force.
 4. Arabian Sea cyclones are increasing due to climate change.
-

- Warm current → More rain
- Cold current → Less rain
- El Niño → Weak monsoon

MEMORY TRICKS

- “Warm = Wet”

- “Cold = Dry”
- “Gulf Stream = Europe warm”
- “Oyashio = Cold Japan side”

DAY 12 – TIDES, WAVES & OCEAN RELIEF

1. WAVES (BASIC CONCEPT)

1. Waves are oscillatory movements of water particles caused mainly by wind blowing over the surface of the ocean.
2. In waves, water particles move in circular paths, but there is no large-scale movement of water from one place to another.
3. The energy of waves is transferred across the ocean surface, not the water itself.
4. The size and strength of waves depend on wind speed, duration, and fetch (distance over which wind blows).

TYPES OF WAVES

1. Wind waves are generated by local winds and are the most common type of waves.
2. Tsunami waves are caused by underwater earthquakes, volcanic eruptions, or landslides, and they have very high energy.
3. Tidal waves are often confused with tsunamis but are actually related to tides caused by gravitational forces.

2. TIDES (VERY IMPORTANT)

1. Tides are the periodic rise and fall of sea levels caused by the gravitational pull of the Moon and the Sun.
2. The Moon has a greater influence on tides than the Sun because it is closer to the Earth.
3. Tides occur twice a day in most coastal areas, resulting in high tide and low tide.

TYPES OF TIDES

Spring Tide

1. Spring tides occur during full moon and new moon when the Sun, Moon, and Earth are aligned.
2. They result in very high high tides and very low low tides.

Neap Tide

1. Neap tides occur during the first and third quarter of the Moon.
2. They result in moderate tidal ranges because the Sun and Moon pull at right angles.

3. OCEAN RELIEF (SEA FLOOR TOPOGRAPHY)

1. The ocean floor is not flat but has varied features like mountains, plains, and trenches.
-

MAJOR FEATURES

Continental Shelf

1. The continental shelf is the shallow, gently sloping part of the ocean floor near the continents.
 2. It is rich in marine resources and supports fishing activities.
-

Continental Slope

1. The continental slope is a steep slope that connects the continental shelf to the deep ocean floor.
-

Abyssal Plains

1. Abyssal plains are deep, flat areas of the ocean floor covered with fine sediments.
-

Ocean Trenches

1. Ocean trenches are the deepest parts of the ocean, formed mainly at convergent plate boundaries.
 2. The Mariana Trench is the deepest known trench in the world.
-

Mid-Ocean Ridges

1. Mid-ocean ridges are underwater mountain ranges formed at divergent plate boundaries.
2. They are sites of new crust formation due to volcanic activity.

4. IMPORTANT CONCEPTS (EXAM BOOSTER)

1. Tsunamis are not tides but are caused by seismic activity under the ocean.
 2. Spring tides have the highest tidal range, while neap tides have the lowest.
 3. Continental shelves are the most productive fishing areas.
 4. Ocean trenches are associated with subduction zones.
-

REVISION (QUICK RECALL)

- Waves → Wind energy
 - Tides → Moon gravity
 - Spring tide → Strong
 - Neap tide → Weak
 - Shelf → Shallow
 - Trench → Deepest
-

MEMORY TRICKS

- “Spring = Strong tide”
- “Neap = Normal/Weak”
- “Shelf = Near land”
- “Trench = Deep drop”

1. PHYSICAL DIVISIONS OF INDIA (CORE STRUCTURE)

1. India is divided into six major physical divisions, which are formed due to geological processes and tectonic movements over millions of years.
2. These divisions include the Himalayan Mountains, Northern Plains, Peninsular Plateau, Indian Desert, Coastal Plains, and Islands.

2. THE HIMALAYAS (VERY IMPORTANT)

1. The Himalayas are young fold mountains formed due to the collision of the Indian Plate with the Eurasian Plate.
2. They act as a natural barrier protecting India from cold winds coming from Central Asia.
3. The Himalayas are divided into three parallel ranges: Himadri, Himachal, and Shiwalik.

HIMALAYAN RANGES

► Himadri (Greater Himalayas)

1. Himadri is the northernmost and highest range with average height above 6000 meters.
2. It contains the highest peaks like Mount Everest and Kanchenjunga.

► Himachal (Lesser Himalayas)

1. Himachal lies south of Himadri and has important hill stations like Shimla, Manali, and Mussoorie.

► Shiwalik (Outer Himalayas)

1. Shiwalik is the southernmost range and consists of loose sediments, making it prone to landslides.

3. NORTHERN PLAINS

1. The Northern Plains are formed by the deposition of alluvial soil by rivers like Ganga, Yamuna, and Brahmaputra.
2. These plains are extremely fertile and support dense population and agriculture.
3. They are divided into Punjab Plains, Ganga Plains, and Brahmaputra Plains.

4. PENINSULAR PLATEAU

1. The Peninsular Plateau is the oldest landmass of India and is composed of hard crystalline rocks.
2. It is divided into Central Highlands and Deccan Plateau.
3. The plateau is rich in minerals such as coal, iron, and manganese.

5. INDIAN DESERT (THAR DESERT)

1. The Thar Desert lies in western India, mainly in Rajasthan, and receives very low rainfall.
2. It is characterized by sand dunes and sparse vegetation.

6. COASTAL PLAINS

1. India has two coastal plains: Western Coastal Plain and Eastern Coastal Plain.

Western Coastal Plain

1. It is narrow and lies between the Western Ghats and the Arabian Sea.
2. It has features like lagoons and backwaters (important in Kerala).

Eastern Coastal Plain

1. It is wider and lies between the Eastern Ghats and the Bay of Bengal.
 2. Major rivers form deltas here, such as the Ganga and Godavari.
-

7. ISLANDS OF INDIA

1. India has two main island groups: Andaman & Nicobar Islands and Lakshadweep Islands.
 2. Andaman & Nicobar Islands are located in the Bay of Bengal and are of volcanic origin.
 3. Lakshadweep Islands are located in the Arabian Sea and are coral islands.
-

8. MAPPING PRACTICE (VERY IMPORTANT)

 You must practice marking these daily:

1. Himalayan ranges (Himadri, Himachal, Shiwalik)
 2. Major rivers (Ganga, Brahmaputra, Narmada, Godavari)
 3. Plateau regions (Deccan Plateau)
 4. Coastal plains (Eastern & Western)
 5. Desert region (Thar Desert)
-

IMPORTANT CONCEPTS (EXAM BOOSTER)

1. The Himalayas are young fold mountains, while the Peninsular Plateau is an old stable landmass.
 2. Northern Plains are formed by alluvial deposits, making them highly fertile.
 3. Western coastal plains are narrow, while eastern coastal plains are wider.
 4. Lakshadweep islands are coral, while Andaman islands are volcanic.
-

REVISION (QUICK RECALL)

- Himalayas → Young fold
 - Plains → Fertile
 - Plateau → Old rocks
 - Desert → Dry
 - Coast → East wide, West narrow
 - Islands → Coral vs volcanic
-

MEMORY TRICKS

- “HPPIIC” → Himalaya, Plains, Plateau, Indian Desert, Islands, Coast
- “West = Narrow, East = Wide”
- “Lakshadweep = Coral”

1. CLIMATE OF INDIA (BASIC UNDERSTANDING)

1. India has a tropical monsoon type of climate, which is characterized by seasonal reversal of winds and distinct wet and dry periods.
 2. The climate of India is influenced by several factors such as latitude, altitude, distance from the sea, relief features, and monsoon winds.
 3. The Tropic of Cancer divides India into tropical and subtropical regions, influencing temperature distribution.
 4. Coastal areas experience moderate climate due to the influence of the sea, while interior regions have extreme climate.
-

2. FACTORS AFFECTING INDIAN CLIMATE (VERY IMPORTANT)

1. Latitude affects solar radiation, with southern India receiving more direct sunlight compared to northern India.
 2. Altitude affects temperature, as higher regions like the Himalayas experience colder conditions.
 3. Distance from the sea causes continentality, leading to extreme temperatures in inland regions.
 4. Relief features like the Himalayas block cold winds from Central Asia and influence rainfall patterns.
 5. Ocean currents also influence coastal climate by modifying temperature and humidity.
-

3. MONSOON IN INDIA (REVISION + ADVANCED)

1. The Indian monsoon is driven by differential heating of land and sea, creating pressure differences.
2. During summer, low pressure develops over northern India, attracting moist winds from the Indian Ocean.

3. These winds bring heavy rainfall and are known as the Southwest Monsoon.
 4. During winter, high pressure develops over land, and dry winds blow from land to sea, known as the Northeast Monsoon.
-

4. SEASONS IN INDIA

Winter Season (Dec–Feb)

1. During winter, northern India experiences cold weather due to high pressure over land.
 2. Some rainfall occurs in northwest India due to western disturbances.
-

Summer Season (Mar–May)

1. Summer is characterized by high temperatures and low pressure over land.
 2. Hot winds like Loo blow in northern India.
-

Rainy Season (Jun–Sep)

1. The Southwest Monsoon brings most of the rainfall to India.
 2. Rainfall distribution is uneven, with some areas receiving heavy rainfall and others less.
-

Retreating Monsoon (Oct–Nov)

1. The monsoon withdraws from India during this period.
 2. Tamil Nadu receives rainfall due to the Northeast Monsoon.
-

5. MONSOON CHARACTERISTICS (VERY IMPORTANT)

1. Monsoon rainfall is seasonal and concentrated in a few months.
2. It is unevenly distributed, leading to floods in some areas and droughts in others.

3. Monsoon has breaks, which are short dry periods during the rainy season.

6. RAINFALL TYPES IN INDIA

1. Orographic rainfall occurs when moist air is forced to rise over mountains like the Western Ghats.
 2. Cyclonic rainfall occurs due to low-pressure systems like cyclones.
 3. Convectional rainfall occurs due to heating of land and rising air.
-

REVISION (QUICK RECALL)

- Climate → Tropical monsoon
- SW Monsoon → Main rain
- NE Monsoon → Tamil Nadu rain
- Himalayas → Barrier
- Western Ghats → Orographic rain

DAY 16 – RIVERS OF INDIA + MAPPING IN

1. CLASSIFICATION OF RIVERS (CORE CONCEPT)

1. Rivers in India are broadly classified into Himalayan rivers and Peninsular rivers based on their origin and flow characteristics.
 2. Himalayan rivers are perennial in nature because they are fed by both glaciers and rainfall throughout the year.
 3. Peninsular rivers are mainly rain-fed and seasonal, showing fluctuations in water flow.
-

7. IMPORTANT CONCEPTS (EXAM BOOSTER)

1. The Himalayas play a crucial role in blocking cold winds and forcing monsoon winds to shed rainfall.
 2. Western Ghats cause heavy rainfall on the windward side and create a rain shadow region on the leeward side.
 3. Tamil Nadu receives rainfall mainly from the Northeast Monsoon.
 4. Rajasthan receives low rainfall due to its location and lack of moisture-laden winds.
-

MEMORY TRICKS

- “SW = Summer rain”
- “NE = Tamil Nadu rain”
- “West Ghats = Heavy rain side”

2. HIMALAYAN RIVERS (VERY IMPORTANT)

1. Himalayan rivers originate from glaciers in the Himalayas and flow through deep valleys before entering the plains.
 2. These rivers form large floodplains and are known for meandering and forming deltas.
-

MAJOR HIMALAYAN RIVER SYSTEMS

Ganga River

1. The Ganga originates from the Gangotri Glacier in Uttarakhand.

2. It flows through northern India and empties into the Bay of Bengal forming a large delta.
3. It is the most important river for agriculture and population in India.

Brahmaputra River

1. The Brahmaputra originates in Tibet as the Tsangpo River.
2. It enters India through Arunachal Pradesh and flows through Assam.
3. It forms one of the largest river systems in the world.

Indus River

1. The Indus originates near Lake Mansarovar in Tibet.
2. It flows through India and Pakistan into the Arabian Sea.
3. It has major tributaries like Jhelum, Chenab, Ravi, Beas, and Sutlej.

3. PENINSULAR RIVERS (VERY IMPORTANT)

1. Peninsular rivers originate from the Peninsular Plateau and flow over hard rocks, making them less meandering.
2. These rivers are generally shorter and have fixed courses compared to Himalayan rivers.

MAJOR PENINSULAR RIVERS

Godavari River

1. The Godavari is the longest river of Peninsular India and is often called the “Dakshin Ganga.”
2. It flows eastward into the Bay of Bengal forming a delta.

Krishna River

1. The Krishna originates in the Western Ghats and flows eastward.
2. It is an important river for irrigation in southern India.

Narmada River

1. The Narmada flows westward into the Arabian Sea.
2. It flows through a rift valley, which is a unique feature.

Kaveri River

1. The Kaveri originates in the Western Ghats and flows into the Bay of Bengal.
2. It is important for agriculture in Tamil Nadu and Karnataka.

4. MAPPING PRACTICE (VERY IMPORTANT)

 You must practice marking daily:

1. Origin of major rivers (Gangotri, Western Ghats, etc.)
2. Direction of flow (east vs west flowing rivers)
3. Major tributaries (Yamuna, Chambal, etc.)
4. Delta regions (Ganga-Brahmaputra, Godavari, Krishna)

5. IMPORTANT CONCEPTS (EXAM BOOSTER)

1. Himalayan rivers are perennial, while Peninsular rivers are seasonal.
 2. East-flowing rivers form deltas, while west-flowing rivers form estuaries.
 3. The Ganga-Brahmaputra delta is the largest delta in the world.
 4. Narmada and Tapi are the main west-flowing rivers.
-

REVISION (QUICK RECALL)

- Himalayan → Perennial
 - Peninsular → Seasonal
 - East flow → Delta
 - West flow → Estuary
 - Narmada → Rift valley
-

MEMORY TRICKS

- “Ganga = Glacier”
- “Godavari = Big peninsular river”
- “Narmada = North to West flow”
- “East rivers = Delta makers”

DAY 17 – SOIL, NATURAL VEGETATION & AGRICULTURE IN

1. SOIL IN INDIA (CORE CONCEPT)

1. Soil is the uppermost layer of the Earth’s crust that supports plant growth and is formed through weathering of rocks over time.
 2. Soil formation depends on factors such as climate, parent rock, relief, vegetation, and time.
 3. Indian soils are diverse due to variations in climate and geological conditions across the country.
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TYPES OF SOIL IN INDIA (VERY IMPORTANT)

Alluvial Soil

1. Alluvial soil is formed by the deposition of sediments by rivers and is found mainly in the Northern Plains.
 2. It is highly fertile and suitable for crops like rice, wheat, and sugarcane.
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Black Soil (Regur Soil)

1. Black soil is formed from volcanic rocks and is found in the Deccan Plateau.
 2. It is ideal for cotton cultivation due to its moisture-retaining capacity.
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Red Soil

1. Red soil is formed due to weathering of crystalline rocks and is rich in iron oxide.
 2. It is less fertile and requires fertilizers for cultivation.
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Laterite Soil

1. Laterite soil is formed in regions with high temperature and heavy rainfall.
 2. It is rich in iron and aluminum but poor in nutrients.
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Desert Soil

1. Desert soil is sandy and found in arid regions like Rajasthan.
2. It has low fertility and poor water retention capacity.

2. NATURAL VEGETATION (VERY IMPORTANT)

1. Natural vegetation refers to plant life that grows naturally without human interference.
2. It depends on climate factors like temperature and rainfall.

TYPES OF VEGETATION IN INDIA

Tropical Evergreen Forests

1. These forests are found in regions with heavy rainfall such as the Western Ghats and Northeast India.
2. They are dense forests with trees that remain green throughout the year.

Tropical Deciduous Forests

1. These forests shed leaves during dry seasons and are found in large parts of India.
2. They are also called monsoon forests.

Thorn Forests

1. These forests are found in arid regions with low rainfall.
2. Plants have adaptations like small leaves and deep roots.

Mountain Forests

1. These forests are found in Himalayan regions with variation according to altitude.

3. AGRICULTURE IN INDIA (VERY IMPORTANT)

1. Agriculture is the backbone of the Indian economy and supports a large population.
 2. It depends heavily on monsoon rainfall.
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CROPPING SEASONS

Kharif Crops

1. Kharif crops are grown during the monsoon season (June–October).
2. Examples include rice, maize, cotton, and sugarcane.

Rabi Crops

1. Rabi crops are grown during winter (October–March).
2. Examples include wheat, barley, and mustard.

Zaid Crops

1. Zaid crops are grown in the short season between Rabi and Kharif.
2. Examples include watermelon and cucumber.

4. IMPORTANT CONCEPTS (EXAM BOOSTER)

1. Alluvial soil is the most fertile and widely spread soil in India.
2. Black soil is best suited for cotton cultivation.
3. Tropical deciduous forests are the most widespread vegetation type in India.
4. Agriculture in India is highly dependent on monsoon rainfall.

REVISION (QUICK RECALL)

- Alluvial → Fertile
- Black → Cotton
- Red → Less fertile
- Evergreen → Dense
- Kharif → Rain crops
- Rabi → Winter crops

MEMORY TRICKS

- “Alluvial = All crops”

- “Black = Cotton king”

- “Kharif = Rainy”

- “Rabi = Winter”

DAY 18 – MINERALS, INDUSTRIES & RESOURCES

1. MINERALS IN INDIA (CORE CONCEPT)

1. Minerals are naturally occurring inorganic substances with definite chemical composition and are essential for industrial and economic development.
2. India is rich in a variety of minerals, which are unevenly distributed across different regions due to geological factors.
3. Minerals are classified into metallic minerals, non-metallic minerals, and energy resources.

TYPES OF MINERALS (VERY IMPORTANT)

Metallic Minerals

1. Metallic minerals contain metals and are further classified into ferrous and non-ferrous minerals.
2. Ferrous minerals include iron ore, manganese, and chromite, which are important for steel production.
3. Non-ferrous minerals include copper, bauxite, and lead, which are used in various industries.

Non-Metallic Minerals

1. Non-metallic minerals do not contain metals and include mica, limestone, and gypsum.
2. These minerals are important for industries like cement and construction.

Energy Minerals

1. Energy minerals include coal, petroleum, natural gas, and uranium.
2. These minerals are used for power generation and fuel.

2. MAJOR MINERAL DISTRIBUTION (MAP BASED)

1. Iron ore is mainly found in Odisha, Jharkhand, Chhattisgarh, and Karnataka.
2. Coal is found in Jharkhand, West Bengal, Odisha, and Chhattisgarh.
3. Petroleum is found in Assam, Gujarat, and offshore areas like Mumbai High.
4. Bauxite is found in Odisha, Jharkhand, and Maharashtra.

3. INDUSTRIES IN INDIA (VERY IMPORTANT)

1. Industries are economic activities concerned with the production of goods and services using raw materials and labor.
2. The location of industries depends on factors like availability of raw materials, labor, power, transport, and market.

TYPES OF INDUSTRIES

Primary Industries

1. These industries are based on natural resources such as mining and agriculture.
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Secondary Industries

1. These industries involve manufacturing and processing of raw materials into finished goods.
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Tertiary Industries

1. These industries provide services such as banking, transport, and communication.
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4. MAJOR INDUSTRIES IN INDIA

Iron and Steel Industry

1. The iron and steel industry is the backbone of industrial development in India.
 2. Major steel plants are located in regions rich in iron ore and coal, such as Jamshedpur, Bhilai, and Durgapur.
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Textile Industry

1. The textile industry is one of the oldest industries in India and is based on cotton, jute, and silk.
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REVISION (QUICK RECALL)

- Minerals → Natural resources
 - Iron → Steel industry
 - Coal → Energy
 - Petroleum → Fuel
 - Industry → Raw material + market
-

2. Major centers include Mumbai, Ahmedabad, and Kolkata.
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IT Industry

1. The IT industry is a major contributor to India's economy and is concentrated in cities like Bengaluru, Hyderabad, and Pune.
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5. RESOURCES (BASIC CONCEPT)

1. Resources are anything that can be used to satisfy human needs and have value.
 2. Resources are classified into natural resources, human-made resources, and human resources.
 3. Natural resources are further classified into renewable and non-renewable resources.
-

6. IMPORTANT CONCEPTS (EXAM BOOSTER)

1. Minerals are non-renewable resources and take millions of years to form.
 2. Iron and coal are basic minerals for industrial development.
 3. Industry location depends on raw materials, power, transport, and labor.
 4. Conservation of resources is essential for sustainable development.
-

MEMORY TRICKS

- “Iron + Coal = Industry base”
- “Bauxite = Aluminium”
- “Coal = Power”
- “Mumbai High = Oil”

1. ECOSYSTEM (CORE CONCEPT)

1. An ecosystem is a functional unit of nature where living organisms interact with each other and with their physical environment.
 2. It consists of biotic components (plants, animals, microorganisms) and abiotic components (air, water, soil, sunlight).
 3. Energy flows through an ecosystem in a unidirectional manner, mainly from the Sun to producers and then to consumers.
-

COMPONENTS OF ECOSYSTEM

Producers

1. Producers are green plants that prepare their own food through the process of Photosynthesis using sunlight, carbon dioxide, and water.
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Consumers

1. Consumers are organisms that depend on other organisms for food and are classified into herbivores, carnivores, and omnivores.
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Decomposers

1. Decomposers like bacteria and fungi break down dead organic matter and recycle nutrients back into the ecosystem.
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2. FOOD CHAIN & FOOD WEB

1. A food chain is a linear sequence of organisms where energy is transferred from one level to another.
2. A food web is a network of interconnected food chains that provides stability to the ecosystem.

3. The trophic levels represent different stages in a food chain, such as producers, primary consumers, and secondary consumers.
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3. ENERGY FLOW (VERY IMPORTANT)

1. Energy decreases at each trophic level due to heat loss, following the 10% law.
 2. This means only about 10% of energy is transferred from one trophic level to the next.
 3. Energy flow in an ecosystem is always unidirectional and cannot be recycled.
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4. BIODIVERSITY (VERY IMPORTANT)

1. Biodiversity refers to the variety of life forms on Earth, including plants, animals, and microorganisms.
 2. It is essential for ecosystem stability, human survival, and environmental balance.
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TYPES OF BIODIVERSITY

1. Genetic diversity refers to variation within species.
2. Species diversity refers to the variety of species in a region.

3. Ecosystem diversity refers to different types of ecosystems such as forests, deserts, and wetlands.

5. BIODIVERSITY HOTSPOTS

1. Biodiversity hotspots are regions with high species diversity and high levels of endemism but are under threat.
2. India has four biodiversity hotspots, including the Himalayas and Western Ghats.

6. THREATS TO BIODIVERSITY

GREENHOUSE EFFECT

1. The greenhouse effect is the trapping of heat in the Earth's atmosphere by gases like carbon dioxide and methane.
2. It is a natural process, but excessive greenhouse gases lead to global warming.

IMPACTS OF CLIMATE CHANGE

1. Rising temperatures lead to melting of glaciers and rising sea levels.
2. Extreme weather events such as floods, droughts, and cyclones are increasing.

REVISION (QUICK RECALL)

- Ecosystem → Living + non-living
- Producers → Plants
- 10% law → Energy loss
- Biodiversity → Variety of life
- Climate change → Global warming

1. Habitat loss due to deforestation and urbanization is the biggest threat to biodiversity.
2. Pollution, climate change, and invasive species also reduce biodiversity.
3. Overexploitation of resources leads to extinction of species.

7. CLIMATE CHANGE (VERY IMPORTANT)

1. Climate change refers to long-term changes in temperature, rainfall, and weather patterns.
2. It is mainly caused by the increase in greenhouse gases due to human activities.

3. It affects agriculture, water resources, and biodiversity.

8. IMPORTANT CONCEPTS (EXAM BOOSTER)

1. Ecosystem is the interaction of biotic and abiotic components.
2. Energy flow is unidirectional and decreases at each trophic level.
3. Biodiversity is essential for ecological balance.
4. Climate change is mainly caused by greenhouse gas emissions.

MEMORY TRICKS

- “Producers = Plants”
- “10% energy rule”
- “Greenhouse = Heat trap”
- “Biodiversity = Bio variety”

1. POLLUTION (CORE CONCEPT)

1. Pollution refers to the introduction of harmful substances into the environment, which causes adverse effects on living organisms and ecosystems.
2. It can affect air, water, soil, and even sound levels, leading to various environmental and health problems.

2. AIR POLLUTION (VERY IMPORTANT)

1. Air pollution is caused by harmful gases and particles such as carbon dioxide, sulfur dioxide, nitrogen oxides, and particulate matter.
2. Major sources include vehicles, industries, burning of fossil fuels, and construction activities.
3. It leads to respiratory diseases, global warming, and reduced visibility.

IMPORTANT PHENOMENA

1. Smog is a combination of smoke and fog, often found in urban areas with heavy pollution.
2. Acid rain is caused by the reaction of sulfur dioxide and nitrogen oxides with water vapor, damaging crops and buildings.
3. Ozone layer depletion occurs due to chemicals like chlorofluorocarbons (CFCs), allowing harmful UV radiation to reach Earth.

3. WATER POLLUTION

1. Water pollution is caused by discharge of industrial waste, sewage, and agricultural chemicals into water bodies.
2. It leads to the spread of diseases and affects aquatic life.

3. Eutrophication occurs when excess nutrients cause excessive growth of algae, reducing oxygen levels in water.

4. SOIL POLLUTION

1. Soil pollution occurs due to excessive use of fertilizers, pesticides, and industrial waste dumping.
2. It reduces soil fertility and affects crop productivity.

5. NOISE POLLUTION

1. Noise pollution is caused by excessive sound from vehicles, industries, and urban activities.
2. It affects human health by causing stress, hearing loss, and sleep disturbances.

6. ENVIRONMENTAL ISSUES (VERY IMPORTANT)

1. Deforestation leads to loss of biodiversity, soil erosion, and climate change.
2. Global warming is caused by increased greenhouse gases, leading to rising temperatures.
3. Desertification is the degradation of land in arid regions due to human activities and climate change.
4. Loss of biodiversity affects ecosystem stability and human survival.

7. CONSERVATION (VERY IMPORTANT)

1. Conservation refers to the protection and sustainable use of natural resources.

2. It is essential for maintaining ecological balance and ensuring resources for future generations.
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METHODS OF CONSERVATION

In-situ Conservation

1. In-situ conservation means protecting species in their natural habitat.
 2. Examples include national parks, wildlife sanctuaries, and biosphere reserves.
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Ex-situ Conservation

1. Ex-situ conservation involves protecting species outside their natural habitat.
 2. Examples include zoos, botanical gardens, and seed banks.
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8. IMPORTANT ORGANIZATIONS & AGREEMENTS

1. World Wide Fund for Nature works for wildlife conservation and environmental protection.
 2. United Nations Environment Programme coordinates global environmental activities.
 3. Paris Agreement aims to limit global warming by reducing greenhouse gas emissions.
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9. IMPORTANT CONCEPTS (EXAM BOOSTER)

1. Air pollution is the most common and visible type of pollution.
 2. Eutrophication reduces oxygen levels in water bodies.
 3. Conservation is essential for sustainable development.
 4. In-situ conservation protects species in natural habitat.
-

REVISION (QUICK RECALL)

- Air → Smog, acid rain
 - Water → Eutrophication
 - Soil → Fertility loss
 - Noise → Health impact
 - Conservation → Protection
-

MEMORY TRICKS

- “Air = Smog”
 - “Water = Eutro (algae)”
 - “In-situ = In nature”
 - “Ex-situ = Exit habitat”
-

DAY 24 – NATIONAL PARKS, WILDLIFE SANCTUARIES & BIODIVERSITY MAPPING IN 🇮🇳

1. PROTECTED AREAS IN INDIA (CORE CONCEPT)

1. Protected areas are regions designated for the conservation of wildlife and biodiversity, where human activities are restricted.

2. India has three main types of protected areas: National Parks, Wildlife Sanctuaries, and Biosphere Reserves.

NATIONAL PARKS vs WILDLIFE SANCTUARIES

1. National Parks provide strict protection to ecosystems, and activities like grazing and human interference are highly restricted.
2. Wildlife Sanctuaries allow limited human activities such as grazing and resource collection under regulation.
3. Biosphere Reserves focus on conservation along with sustainable development and research.

2. IMPORTANT NATIONAL PARKS (VERY IMPORTANT + MAP)

Jim Corbett National Park

1. It is the oldest national park in India and is famous for Project Tiger conservation.

Gir National Park

1. It is the only natural habitat of Asiatic lions in the world.

Kaziranga National Park

1. It is famous for the one-horned rhinoceros and is a UNESCO World Heritage Site.

Ranthambore National Park

1. It is known for its tiger population and historical fort inside the park.

Sundarbans National Park

1. It is famous for mangrove forests and the Royal Bengal Tiger.

3. IMPORTANT WILDLIFE SANCTUARIES

Keoladeo National Park

1. It is famous for migratory birds and is a UNESCO World Heritage Site.

Periyar Wildlife Sanctuary

1. It is known for elephants and scenic lake views.

4. BIOSPHERE RESERVES (IMPORTANT)

1. Biosphere reserves are large areas for conservation of biodiversity, including core, buffer, and transition zones.
2. Examples include Nilgiri, Nanda Devi, and Gulf of Mannar biosphere reserves.

5. BIODIVERSITY MAPPING (VERY IMPORTANT)

 Practice marking these daily:

1. Tiger reserves (Corbett, Ranthambore, Sundarbans)
2. Rhino habitat (Kaziranga)
3. Lion habitat (Gir)
4. Elephant reserves (Kerala, Karnataka)
5. Mangrove forests (Sundarbans)

6. IMPORTANT SPECIES & LOCATIONS

1. Asiatic Lion → Gir (Gujarat)
2. One-horned Rhino → Kaziranga (Assam)
3. Royal Bengal Tiger → Sundarbans
4. Elephant → Western Ghats region
5. Snow Leopard → Himalayan region

7. IMPORTANT CONCEPTS (EXAM BOOSTER)

1. National Parks have stricter protection than Wildlife Sanctuaries.
 2. India is a mega-biodiversity country with rich flora and fauna.
 3. Biodiversity hotspots in India include Western Ghats and Himalayas.
 4. Mapping of parks and species is frequently asked in exams.
-

REVISION (QUICK RECALL)

- Corbett → Tiger
 - Gir → Lion
 - Kaziranga → Rhino
 - Sundarbans → Mangrove + Tiger
 - Periyar → Elephant
-

MEMORY TRICKS

- “GLR = Gir Lion Reserve”
- “Kaziranga = Rhino land”
- “Sundar = Sundarbans (Mangrove)”
- “Corbett = First tiger park”